

Soham Jana

Assistant Professor, University of Notre Dame

Department of Applied and Computational Mathematics and Statistics

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Research Interests

Theoretical and methodological aspects of high-dimensional statistics, neural networks, mixture modeling, distance-based estimators.

Education

Ph.D. in Statistics and Data Science, Yale University 2022

Advisor: Yihong Wu

Thesis: Learning non-parametric and high-dimensional distributions via information-theoretic methods

Master of Statistics (First class with distinction), Indian Statistical Institute 2017

Specialization: Theoretical Statistics

Dissertation: Characterization of single-integral non-kernel divergences

Advisor: Ayanendranath Basu

Bachelor of Statistics (First class with distinction), Indian Statistical Institute 2015

Employment

University of Notre Dame 2024–Present

Assistant Professor, Department of Applied and Computational Mathematics and Statistics

Princeton University 2022–2024

Postdoctoral Researcher, Department of Operations Research and Financial Engineering

Hosts: Sanjeev Kulkarni, Jianqing Fan

Researcher, First Republic Bank Research and Lifelong Learning Program 2022–2023

Lecturer Spring 2023, Fall 2023

Grants and Awards

Kaneb Center Course Design Academy, University of Notre Dame 2024–2025

Award Amount: \$5000

Publications

* Authors not in alphabetical order † Equal contribution

Journal Publications

- J1. **Jana, S.**, Yang, K., & Kulkarni, S. (2026).*
Adversarially robust clustering with optimality guarantees.
IEEE Transactions on Information Theory, 72(1), 478–500.
- J2. **Jana, S.**, Polyanskiy, Y., & Wu, Y. (2025).
Optimal empirical Bayes estimation for the Poisson model via minimum-distance methods.
Information and Inference, Volume 14, Issue 4.
- J3. **Jana, S.**, Fan, J., & Kulkarni, S. (2025).*
A provable initialization and robust clustering method for general mixture models.
IEEE Transactions on Information Theory, vol. 71, no. 9.
- J4. **Jana, S.**, Li, H., Yamada, Y., & Lindenbaum, O. (2023).
Support recovery with stochastic gates: theory and application for linear models.
Signal Processing, 213.
- J5. Han, Y., **Jana, S.**, & Wu, Y. (2023).
Optimal prediction of Markov chains with and without spectral gap.
IEEE Transactions on Information Theory, 69(6).
- J6. **Jana, S.**, & Basu, A. (2019).*
A characterization of all single-integral, non-kernel divergence estimators.
IEEE Transactions on Information Theory, 65(12).

Conference Publications

- C1. **Jana, S.**, Polyanskiy, Y., Teh, A., & Wu, Y. (2023).
Empirical Bayes via ERM and Rademacher complexities: the Poisson model.
Conference on Learning Theory (COLT), PMLR 195:5199–5235.
- C2. Han, Y., **Jana, S.**, & Wu, Y. (2021).
Optimal prediction of Markov chains with and without spectral gap.
Neural Information Processing Systems (NeurIPS).
- C3. **Jana, S.**, Polyanskiy, Y., & Wu, Y. (2020).
Extrapolating the profile of a finite population.
Conference on Learning Theory (COLT).

Preprints

- P1. **Jana, S.** (2026).
Minimax Theory of Likelihood-Based Deep Learning for Speckle Regression
arXiv:2607.14064.
- P2. Xing, H.[†], **Jana, S.**[†], & Maleki, A. (2025).*
Minimax analysis of estimation problems in coherent imaging.
arXiv:2508.18503.
- P3. Fan, J., **Jana, S.**, Kulkarni, S., & Yin, Q. (2025).
Factor informed double deep learning for average treatment effect estimation.
arXiv:2508.17136.
- P4. Chen, X.[†], **Jana, S.**[†], Metzler, C., Maleki, A., & Jalali, S. (2025).*
Multilook coherent imaging: theoretical guarantees and algorithms.
Under major revision at *IEEE Transactions on Informaiton Theory* arXiv:2505.23594.

- P5. Tang, S., **Jana, S.**, & Fan, J. (2024).
 Factor adjusted spectral clustering for mixture models.
 arXiv:2408.12564
 Under major revision at *Journal of the American Statistical Association*.

Conferences, Workshops, and Invited Talks

- Conference on Statistical Foundations of Artificial Intelligence. California, USA (November 2026)
- Statistical Learning and Data Science: Inference and Intelligence, New York, USA (November 2026)
- IMSI Workshop on Reinforcement Learning from Offline Data and Human Feedback, Chicago, USA (April 2026)
- Joint Statistical Meetings, Nashville, TN, USA (August 2025)
- IMS New Researchers Conference, Nashville, TN, USA (July–August 2025)
- International Webinar on Recent Trends in Statistical Theory and Applications, Kerala, India (July 2025)
- International Indian Statistical Association, Lincoln, NE, USA (June 2025)
- International Indian Statistical Association, Cochin, Kerala, India (December 2024)
- Joint Statistical Meetings, Portland, OR, USA (August 2024)
- University of Notre Dame Statistics Department Seminar, Notre Dame, IN, USA (February 2024)
- University of Wisconsin–Madison Statistics Department Seminar, Madison, WI, USA (February 2024)
- University of Texas at Dallas Statistics Department Seminar, Richardson, TX, USA (January 2024)
- Indian Statistical Institute ISRU Department Seminar, Kolkata, India (July 2023)
- Conference on Learning Theory (COLT), Bangalore, India (July 2023)
- Neural Information Processing Systems (NeurIPS), Virtual (December 2021)
- Conference on Learning Theory (COLT), Graz, Austria (July 2020)

Peer-Review and Related Activities

Paper Reviews:

- Annals of Statistics (4)
- IEEE Transactions on Information Theory (5)
- Journal of the American Statistical Association (2)
- Neural Information Processing Systems – NeurIPS (4)
- Algorithmic Learning Theory (7)
- Electronic Journal of Statistics (2)
- IEEE International Symposium on Information Theory (1)
- Journal of the Royal Statistical Society (2)
- IEEE Transactions on Image Processing (1)
- IEEE Wireless Communications Letters (1)
- Bernoulli (1)
- Statistica Sinica (1)

- Stat – an ISI Journal (1)
- Journal of Statistical Planning and Inference (1)
- Biometrics (1)
- SIAM Journal on Mathematics of Data Science (1)

Session Organization and Chairing:

CFE-CMStatistics Conference	December 2026
Session organizer: <i>Future directions in computational imaging and machine learning</i>	
CFE-CMStatistics Conference	December 2025
Session organizer: <i>Recent Advances in Causal Inference</i>	
Joint Statistical Meetings	August 2025
Session chair: <i>New Advances in Optimization Algorithms for Causal Discovery</i>	
Joint Statistical Meetings	August 2024
Session chair: <i>New Advances in Nonparametric Hypothesis Testing – Part I</i>	
Joint Statistical Meetings	August 2024
Session chair: <i>New Developments in Non-Euclidean Statistics</i>	
IEEE Conference on Information Sciences and Systems	March 2024
Session chair: <i>Machine Learning and Statistical Inference</i>	

Admissions and Evaluation Committees:

Yale S&DS M.A. Admission Committee, Reviewer	2021
One of the committee members handling over 150 applications and making admission recommendations.	

Teaching and Advising

Brief Overview of Undergraduate and Graduate Teaching

I have taught undergraduate probability and graduate-level machine learning courses at the University of Notre Dame (since Fall 2024), and probability and statistical machine learning at Princeton University (2023). My courses range from foundational probability for undergraduates to graduate level courses and seminars on high-dimensional methods and machine learning.

Regular Courses Taught

University of Notre Dame

ACMS 30530: Introduction to Probability	Fa2024, Fa2025, Sp2026
ACMS 70100: Machine Learning Methods for High-dimensional Data	Sp2026
ACMS 80870: Modern Machine Learning Techniques with Application	Sp2025
ACMS-67695-07 Directed Master's Project	Sp2025

Princeton University

ORF 309/ENG 309/MAT 380: Probability and Stochastic Systems	Sp2023
ORF 570: Statistical Machine Learning	Fa2023

PhD Students Supervised

1. Geethu Rajan (April 2026 – current)

Undergraduate Students Supervised

1. Michael Budd, Summer undergraduate research program (June 2026 – current)

Department, College, and University Service

Abbreviations spelled out: COSE = Colleges of Science and Engineering; DAC = Data, AI, and Computing; ASA = American Statistical Association.

Committee member for Ph.D. candidacy of ACMS student (Zhehao Liang)	Sp2026
ASA DataFest at Notre Dame – organized by the ACMS Department	Sp2026 – present
Served as a mentor in this undergraduate data analysis competition	
ACMS Department – Statistics seminar chair	August 2025 – Present
COSE Research Horizons Poster Presentation Judging	December 2025
DAC Working Group on AI Foundations – Served in the core committee	2025 – 2026
Responsibilities: Identify funding opportunities, helped organize AI Foundations mini symposium and moderated the panel “Challenges and Opportunities in AI Foundations” there.	
Reader for Ph.D. Defense (for Dr. Shivam Kumar)	2025

Professional Affiliations, Service, and Leadership

Organizational Membership and Service:

Institute of Mathematical Statistics (IMS)
 American Statistical Association (ASA)
 International Indian Statistical Association (IISA)

Yale University Service:

Yale S&DS Graduate Reading Group, Co-organizer	2020
Scheduled talks and led discussion sessions.	
Yale Women in Data Science (WiDS) Workshop, Mentor	2020
Served as mentor for Yale undergraduate students participating in the WiDS Datathon Challenge 2020.	
South Asian Graduate and Professional Association at Yale (SAGA)	2018 – 2021
Treasurer, core committee member, and cultural committee head.	
Organized socio-cultural events to promote cultural exchanges at Yale.	

Major Consultancies in the Public and Private Sectors

Researcher, The First Republic Bank Research and Lifelong Learning Program, Princeton University	June 2022 – May 2023
Conducted research on statistical and machine learning methods in a banking and financial research context.	

Community Work

Teaching at Math Circle, University of Notre Dame Spring 2025, Fall 2025
Promoting STEM education among school children in the local South Bend community.

Other Awards

INSPIRE Scholarship, Government of India (2012–2017)
INMO Merit Certificate (Top 75, 2012)